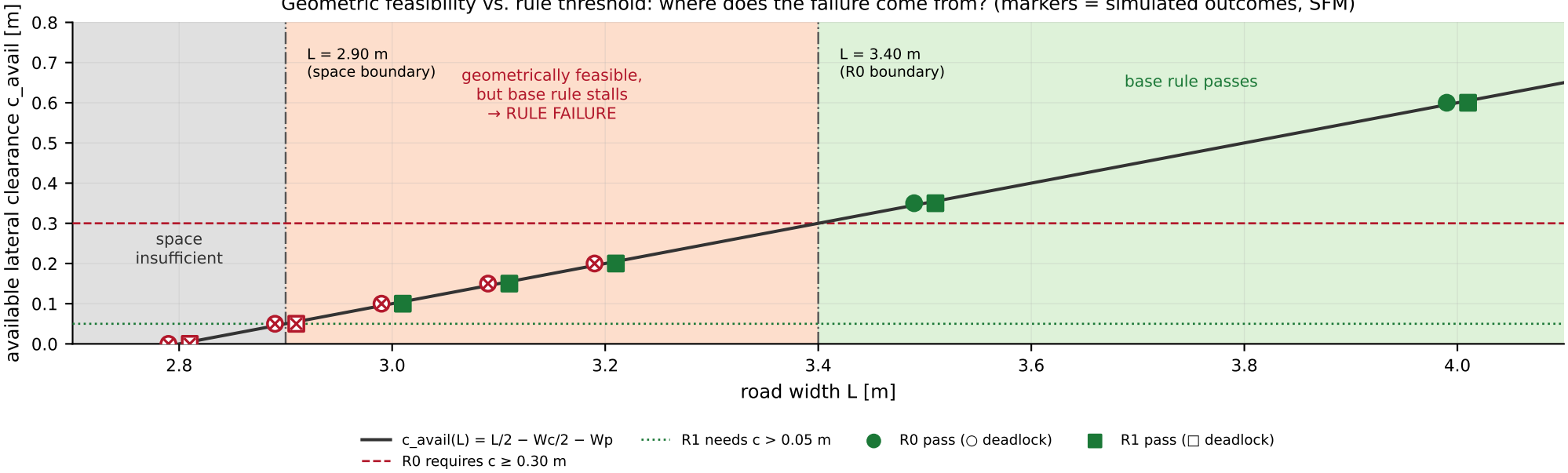
Geometric feasibility vs. rule threshold: where does the failure come from? (markers = simulated outcomes, SFM)
available lateral clearance c_avail [m]
road width L [m]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
2.8
3.0
3.2
3.4
3.6
3.8
4.0
L = 2.90 m
(space boundary)
L = 3.40 m
(R0 boundary)
space
insufficient
geometrically feasible,
but base rule stalls
→ RULE FAILURE
base rule passes
c_avail(L) = L/2 − Wc/2 − Wp
R0 requires c ≥ 0.30 m
R1 needs c > 0.05 m
R0 pass (○ deadlock)
R1 pass (□ deadlock)